

HackTheBox Academy

Network Enumeration with Nmap module

Course: Junior Cybersecurity Associate

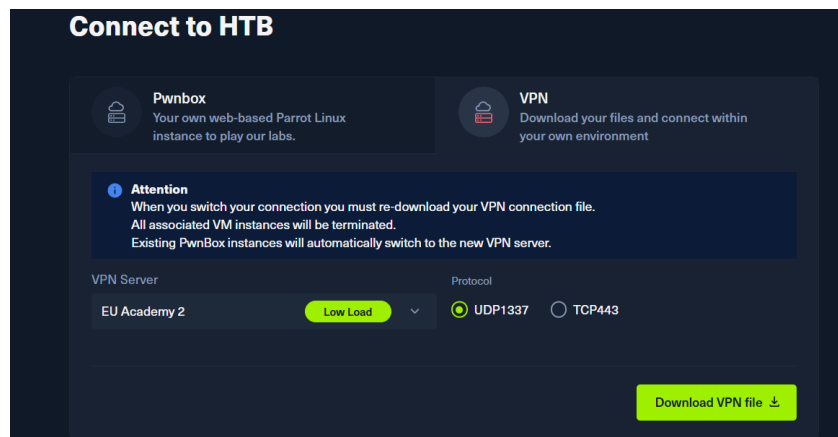
Host and Port scanning.

Nmap is an open source scanning tool used to scan networks and identify which hosts are available on the network using raw packets, services and application.

In this section we were given two tasks:

- Find all TCP ports on a target and submit the total open ports found;
- Enumerate the hostname on the target and submit it as the answer.

Before completing the tasks we first connect to HackTheBox's lab network vpn.



In order to do that we download the VPN file in the lower section of the topic's page

After downloading we use a command: **sudo openvpn <file name>.ovpn**

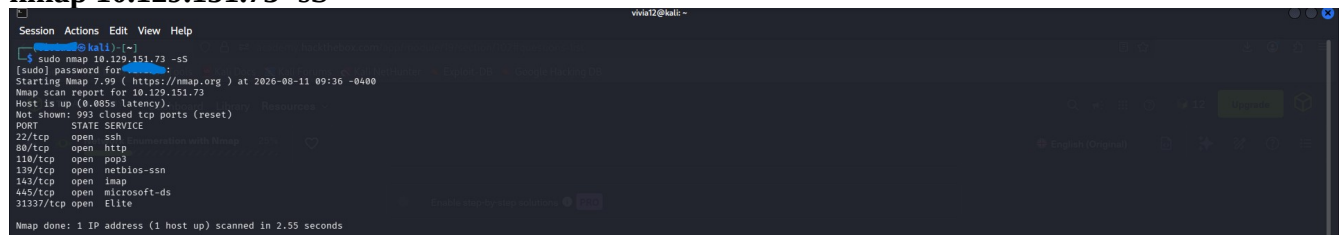
The command will open the vpn and we know we are connected if it says: **Initialization Sequence Complete**

1.Find all TCP ports on a target and submit the total open ports found.

We first spawn HackTheBoxe's target IPv4 address which was: **10.129.151.73**

This is the address we will use for our tasks.

So in order to find all TCP ports on the 10.129.151.73 IPv4 address we will type a command: **sudo nmap 10.129.151.73 -sS**



```
Session Actions Edit View Help
vivia12@kali: ~
$ sudo nmap 10.129.151.73 -sS
[sudo] password for vivia12:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:36 -0400
Nmap scan report for 10.129.151.73
Host is up (0.085s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
110/tcp   open  pop3
139/tcp   open  netbios-ssn
143/tcp   open  imap
445/tcp   open  microsoft-ds
31337/tcp open  Elite
Nmap done: 1 IP address (1 host up) scanned in 2.55 seconds
```

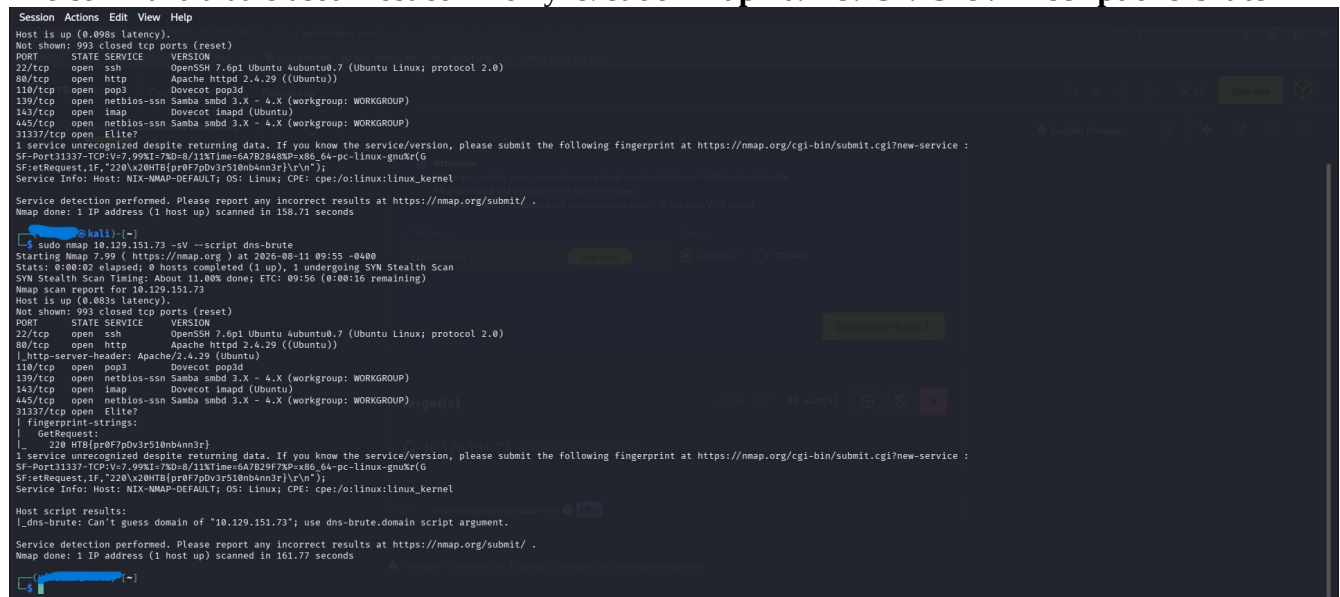
-sS is a SYN scan that scans the first one thousand ports.

In the screenshot we see that the answer is 7, because there are 7 open ports, that the scan has found.

2. Enumerate the hostname on the target and submit it as the answer.

In order to enumerate the hostname on the target, we will use Nmap's **-sV** option, that is used to get additional information from those ports as it does a reverse DNS lookup and retrieves information such as hostname, service names and other details about the target. During this section I was unsure about the command which is correct so this command was found on a Medium write up by: **H3rCul3s**

The command that is used most commonly is: **sudo nmap 10.129.151.73 -sV - - script dns-brute**



```
Session Actions Edit View Help
vivia12@kali: ~
$ sudo nmap 10.129.151.73 -sV --script dns-brute
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400
Stats: 0:00:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 11.00% done; ETC: 09:56 (0:00:16 remaining)
Nmap scan report for 10.129.151.73
Host is up (0.083s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.6p1 Ubuntu Aubuntu0.7 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.29 ((Ubuntu))
110/tcp   open  pop3         Dovecot pop3d
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
143/tcp   open  imap         Dovecot imapd (Ubuntu)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
31337/tcp open  Elite
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port31337-TCP:V=7.99RI=7ND=8/11KTime=6A7B2040Bpx86_64-pc-linux-gnuKr(G
SF-SetRequest.IF:"220x20HTB[pr0f7p0v3r510nbann3r]\r\n");
Service Info: Host: NIX-NMAP-DEFAULT; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 158.71 seconds

vivia12@kali: ~
$ sudo nmap 10.129.151.73 -sV --script dns-brute
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400
Stats: 0:00:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 11.00% done; ETC: 09:56 (0:00:16 remaining)
Nmap scan report for 10.129.151.73
Host is up (0.083s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.6p1 Ubuntu Aubuntu0.7 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.29 ((Ubuntu))
110/tcp   open  pop3         Dovecot pop3d
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
143/tcp   open  imap         Dovecot imapd (Ubuntu)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
31337/tcp open  Elite
1 fingerprint-strings:
|_  GetRequest:
|_  220 HTB[pr0f7p0v3r510nbann3r]
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port31337-TCP:V=7.99RI=7ND=8/11KTime=6A7B2040Bpx86_64-pc-linux-gnuKr(G
SF-SetRequest.IF:"220x20HTB[pr0f7p0v3r510nbann3r]\r\n");
Service Info: Host: NIX-NMAP-DEFAULT; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ dns-brute: Can't guess domain of "10.129.151.73"; use dns-brute.domain script argument.

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 161.77 seconds

vivia12@kali: ~
$
```

--script – Nmap during/after the scan runs a particular NSE script to obtain more information.

Dns-brute – a particular script that tries to discover DNS hostnames/subdomains.

So the answer is: NTX-NMAP-DEFAULT